

A black bird with a red eye is perched on a branch. The bird has a long, dark beak and its feathers have a subtle iridescent sheen. The background is a soft, out-of-focus green.

Foraging Ecology

IB 431

Agenda

- Go over methods assignment.
- General foraging tactics.
- Discuss Flower 2013.

Methodology

- Explain protocol (most of you have started this in some capacity)
- Define treatment groups.
- Identify study species and study site
- Clearly identify measured variables (and how)
- 20 potential points

Negative Control B.

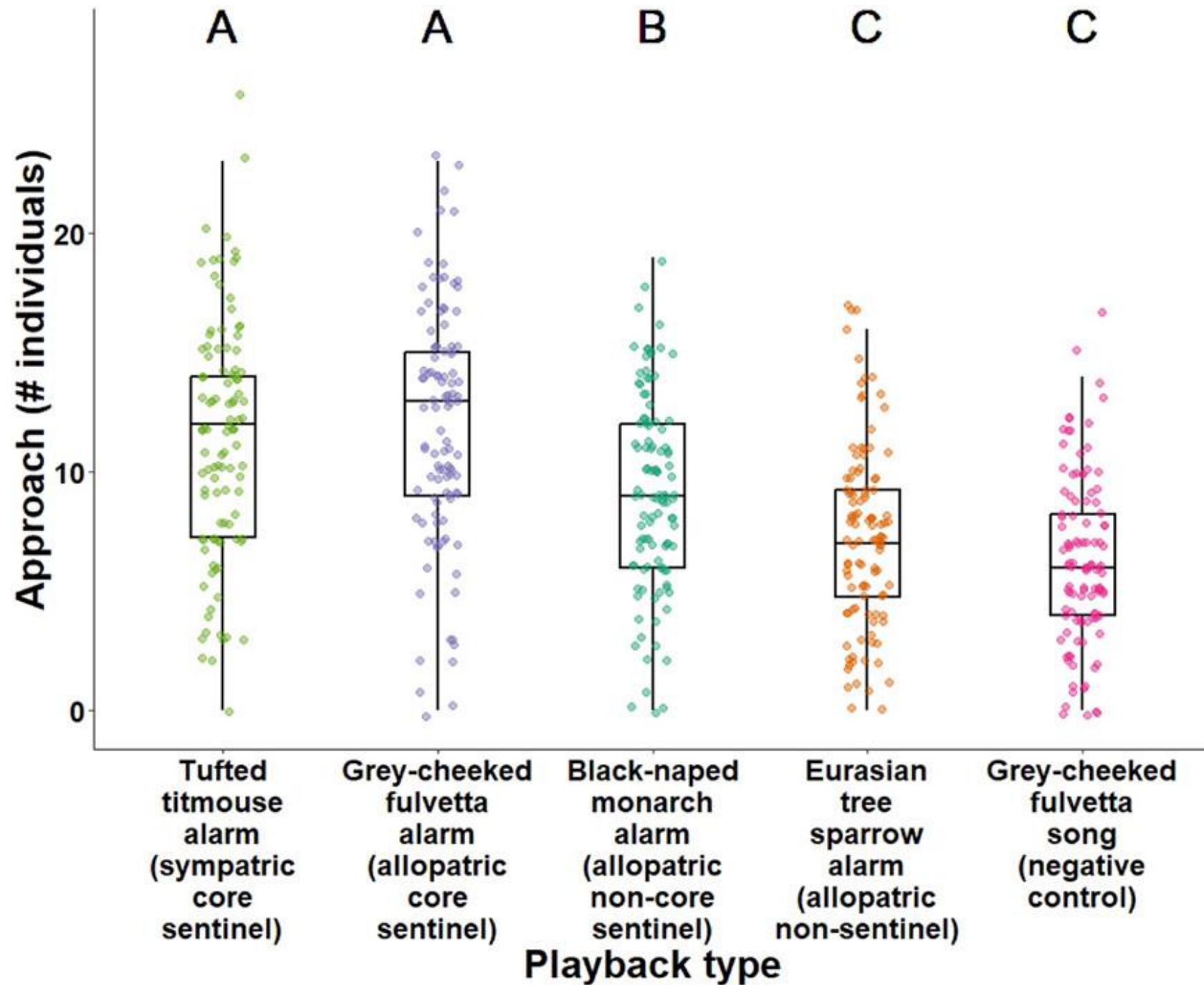
The Kalij Pheasant (*Lophura leucomelanos*) is an introduced species native to India. This species does not participate in mixed-species flocks and does not produce alarm calls for heterospecifics. This species occurs on Oahu but should not elicit a response from heterospecifics. As a result, we shall play its alarm call as an alarm call that should receive no response, regardless of learned history, evolutionary history, or flocking experience.

Field Protocol

To test responses of mixed species flock members to various sentinel calls, we will conduct playbacks in Hawaii in areas where both native and introduced species occur. Each trial will be fifteen minutes in total duration, consisting of three phases, with a five-minute pre-period, five minutes of playback, and a five-minute post-period. During the first trial at a given site, we will estimate the 15-meter mark from the speaker and the 50-meter mark. All bird activity within 50-meter distance from the speaker will be recorded including distance from speaker, vocalizations, and height in vegetation for each individual bird present (Matsuoka et al. 2014, Martínez et al. 2018). Data will be continuously collected via audio observations recorded into a voice recorder and then later transcribed. Each site will be visited five times, with each treatment type being presented at that site once. Each site will have all trials conducted at it within a single day, with a minimum of one hour between trials to prevent birds from growing overstimulated. We aim to collect data at 100 sites over the course of a 40-day sampling period.

Figure

- Identify measured variables.
- Look at papers you're following up for ideas
- (Hand-drawn fine)
- 10 points



Captioning Figures

Figure 2

[Open in figure viewer](#)

[↓ PowerPoint](#)

Per-trial approach of birds in response to the vocalizations of a variety of birds. Letters indicate statistically different groups, box plots indicate 10, 25, 50, 75 and 90th percentiles, and the dots are all the data points (n = 538 trials)

- Descriptive explanation of figure (e.g. number of shared feeding over time by male vampire bats).
- List sample size.

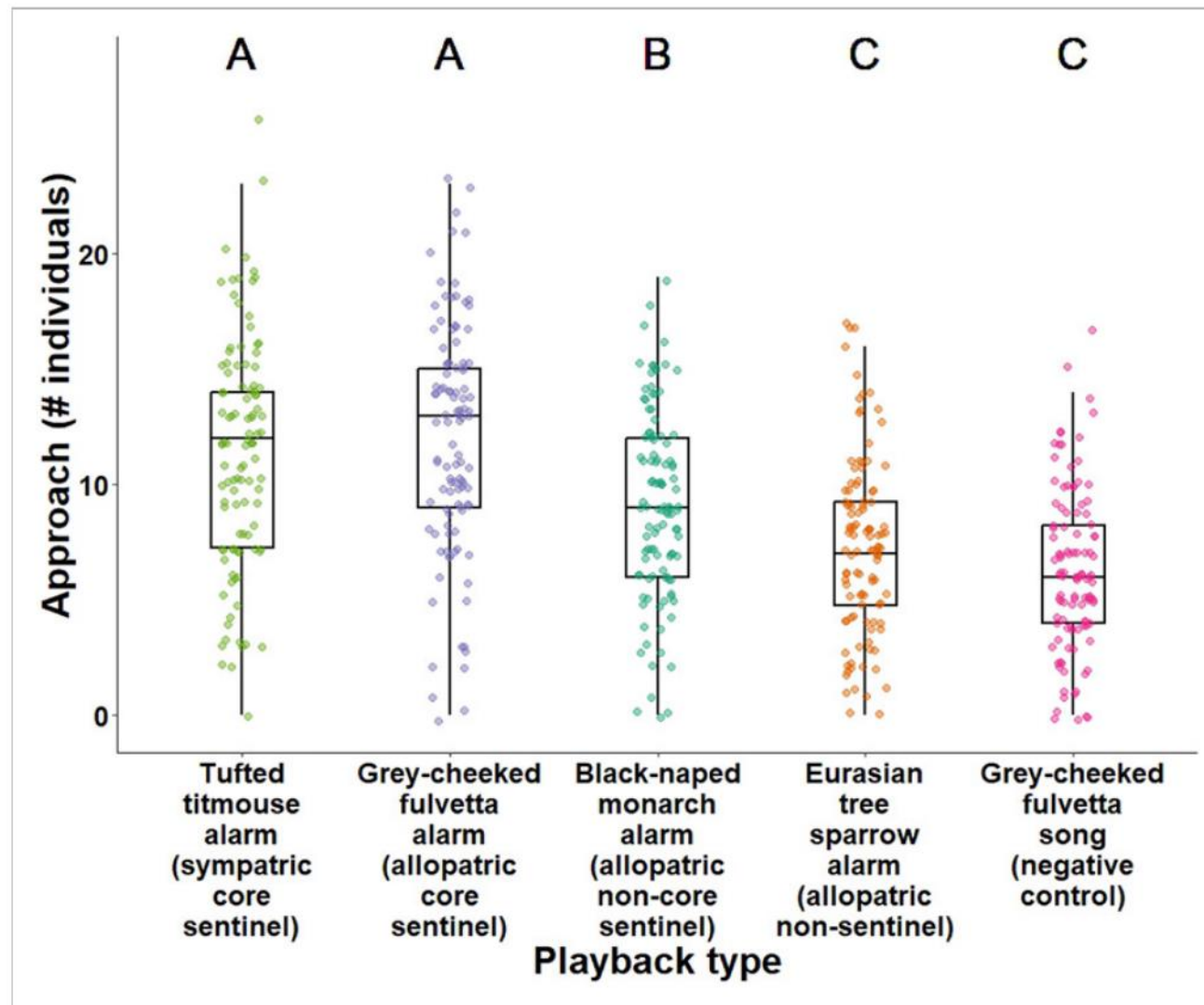


Figure 2

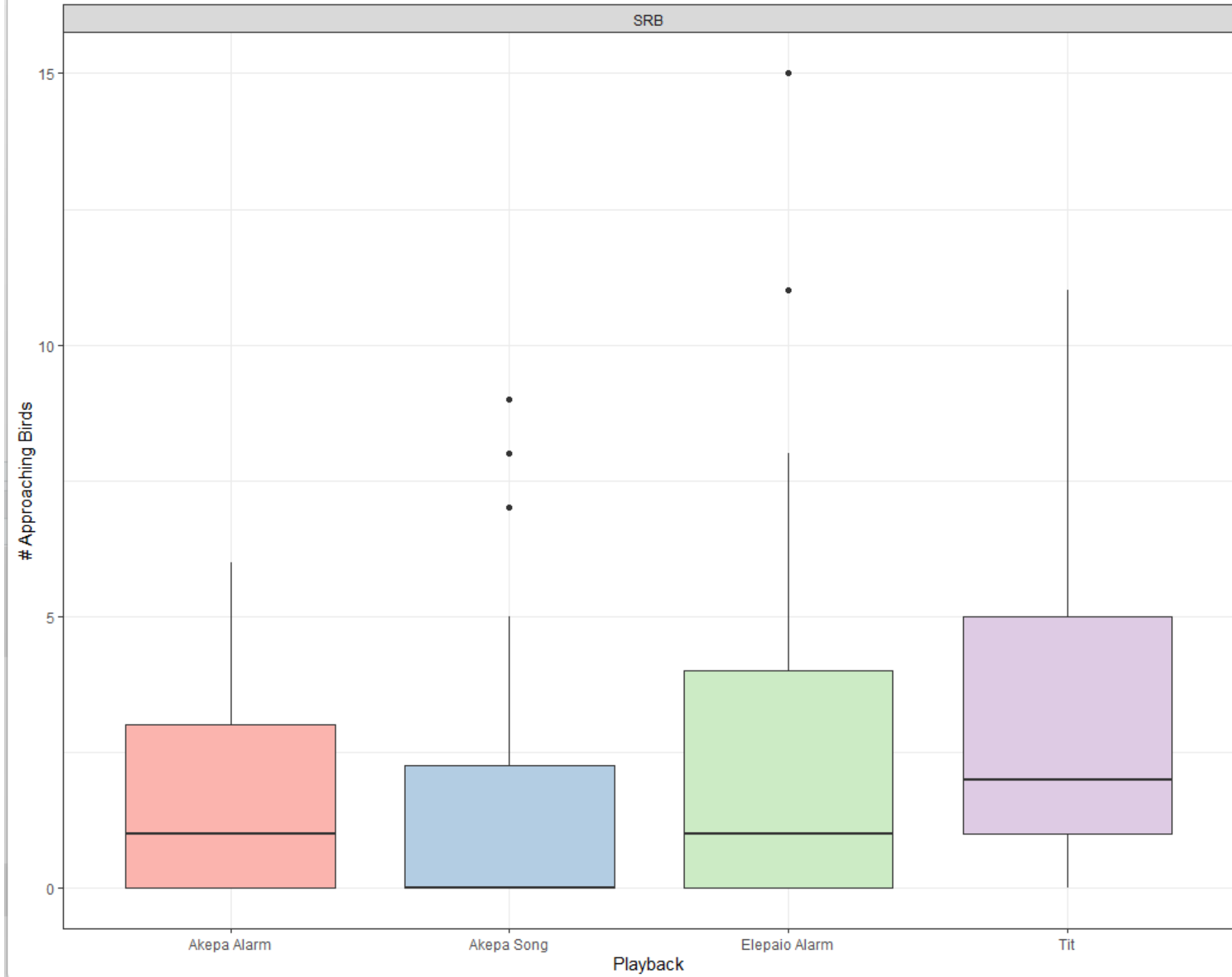
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Per-trial approach of birds in response to the vocalizations of a variety of birds. Letters indicate statistically different groups, box plots indicate 10, 25, 50, 75 and 90th percentiles, and the dots are all the data points (n = 538 trials)

Alternate Outcomes

- What if you're wrong?
- What does it mean?
- Contrast with original hypotheses
- 10 potential points



Foraging Writ Large

The act of searching for food resources.

Highly varied.

Many complex strategies.

Series of constant choices.

- Stay or Go?
- Alone or Together?
- Large or Small Prey?

Some choices are flexible, others evolutionarily ingrained.



Patches

A clump of resources.

- A berry bush.
- A field of grain.
- A group of insects.
- A river where wildebeest frequent.

Typical foraging theory predicts that animals will occupy a patch until it is no longer profitable.

Patches typically depleted over time, either by individual or con- and heterospecifics.



Incomplete Information

Tradeoffs (again).

Choosing to forage in one location inevitably means you give up on another.

How do you know when to leave a patch?

Typically, when net gain is no longer positive.



Incomplete Information

Animals do not have the full picture at all times.

Foraging involves balancing potential future payout with current options.

Use previous experiences to help inform decision making.

You can leave your current patch for a new patch that *might* be better, but what are the odds that it'll be good?

If you're wrong, what are the costs?



Steps to Foraging

While foraging is treated as a singular act, made up of multiple component parts.

Complexity of steps is species-dependent.

Different species have different costs to each.

Search.

Active time spent foraging without encountering prey item.

Encounter.

Point when a prey item is encountered and an animal decides whether to consume or abandon.

Handle.

Time and energy to pursue, capture and consume prey item.

$$E(h_r) = \mu_h = [\sum \lambda_i h_i^*] / \lambda^* \quad (1.4)$$

$$E(\xi_r) = \mu_e = [\sum \lambda_i e_i^*] / \lambda^* \quad (1.5)$$

$$V(\xi_r) = \sigma_e^2 = [\sum \lambda_i E(e_i^2)] / \lambda^* - \mu_e^2 \quad (1.6)$$

$$V(h_r) = \sigma_h^2 = [\sum \lambda_i E(h_i^2)] / \lambda^* - \mu_h^2 \quad (1.7)$$

$$\begin{aligned} Cov(T_r, \xi_r) &= \sigma_{re} = Cov(h_r + T_{sr}, \xi_r) \\ &= Cov(h_r, \xi_r). \end{aligned} \quad (1.8)$$

Now:

$$T_r = T_{sr} + h_r, \quad \text{so} \quad \mu = E[T_{sr}] + \mu_h$$

or

$$\mu = 1/\lambda^* + [\sum \lambda_i h_i] / \lambda^*. \quad (1.9)$$

Since

$$\begin{aligned} Cov(T_{sr}, \xi_r) &= 0, \quad \sigma^2 = V[T_{sr}] + \sigma_h^2 \\ \sigma^2 &= 1/\lambda^{*2} + \sigma_h^2. \end{aligned} \quad (1.10)$$

If we substitute (1.5) and (1.9) into (1.1) the following results as $\tau \rightarrow \infty$.

$$\frac{\mu_\tau}{\tau} \approx \frac{\sum \lambda_i e_i^*}{1 + \sum \lambda_i h_i^*} \quad (1.11)$$

substituting (1.5) and (1.10) into (1.2) gives the variance for G_τ as $\tau \rightarrow \infty$.

$$\begin{aligned} \sigma_\tau^2 &\approx \tau \left[\frac{\sigma_e^2}{D} + \frac{\mu_e^2 (1 + \sigma_h^2 \lambda^{*2}) \lambda^*}{D^3} - \frac{2\mu_e \sigma_{re}}{D^2} \right] \\ \text{where } D &= 1 + \sum \lambda_i h_i^*. \end{aligned} \quad (1.12)$$

Searching

Active

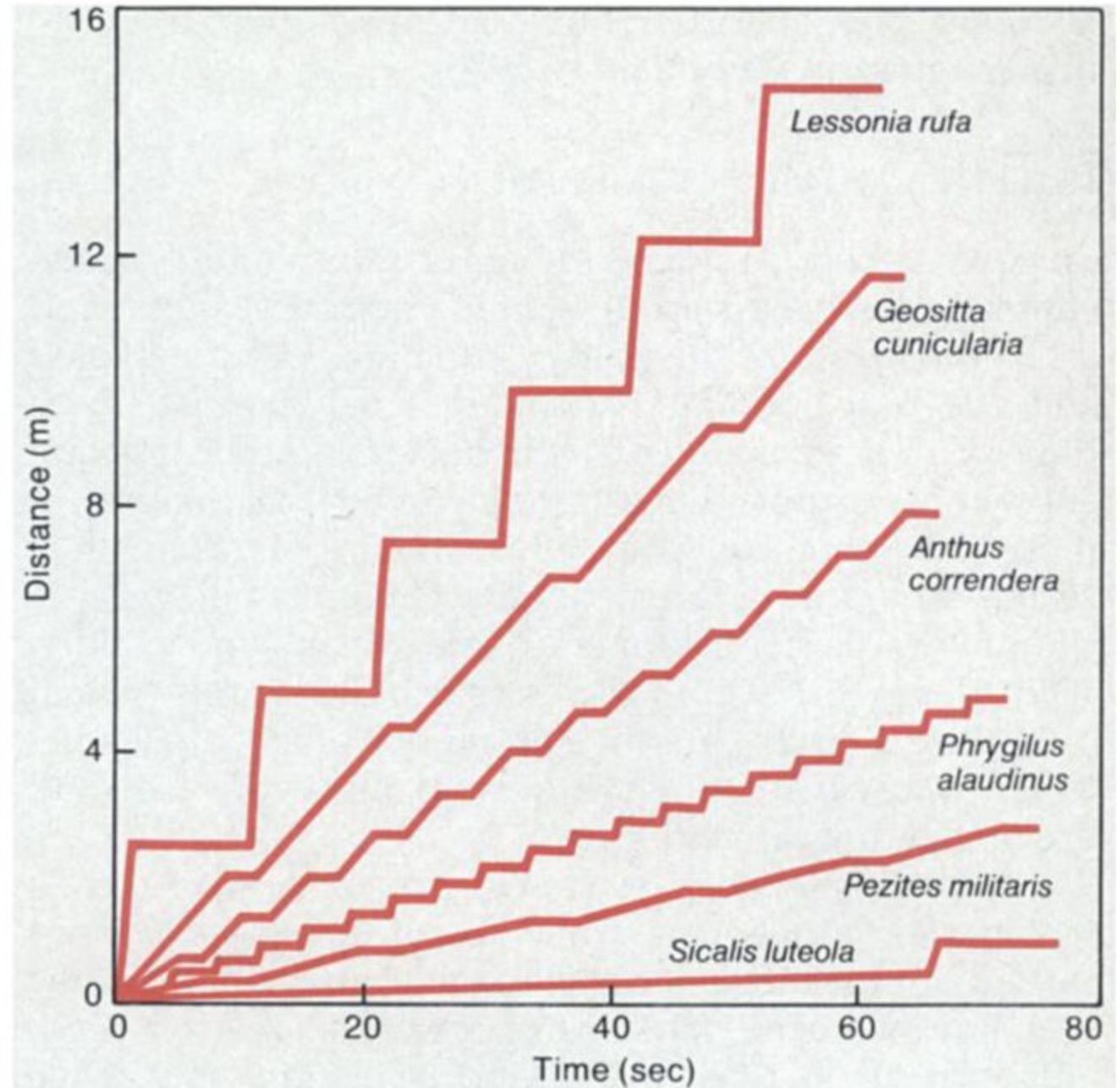
- Continuous
- Active, Few Rests

Ambush

- Lie in Wait
- Sedentary, Bursts of Movement

This is a spectrum– all animals move in some capacity. All animals rest in some capacity.

O'brien, W. John, Howard I. Browman, and Barbara I. Evans.
"Search strategies of foraging animals." *American Scientist*
78.2 (1990): 152-160



Active Foraging

Seek out prey.

Prey tends to be more sedentary.

Expend high amounts of energy but can cross distances in response to fluctuating resources.

Higher activity increases risk of predation.

Huey, R.B. and Pianka, E.R. (1981), Ecological Consequences of Foraging Mode. *Ecology*, 62: 991-999. <https://doi-org.proxy2.library.illinois.edu/10.2307/1936998>



Ambush Foraging

Sedentary foraging strategy.

Rely on prey to come to you.

Active prey.

Low energy, low risk.

Requires high prey density (or at least enough to survive).



Prey Type Influences Searching Strategy

Suppose you are a predator– you are hunting a prey that exists in large quantities but exists in clumps (e.g. insects in specific grassland areas). Which search strategy would benefit you?

What if you are in a high-activity area (e.g. a coral reef), where many fish come through, but are unlikely to stick to any one place?

What if you are a predator in a vast space with low prey density?

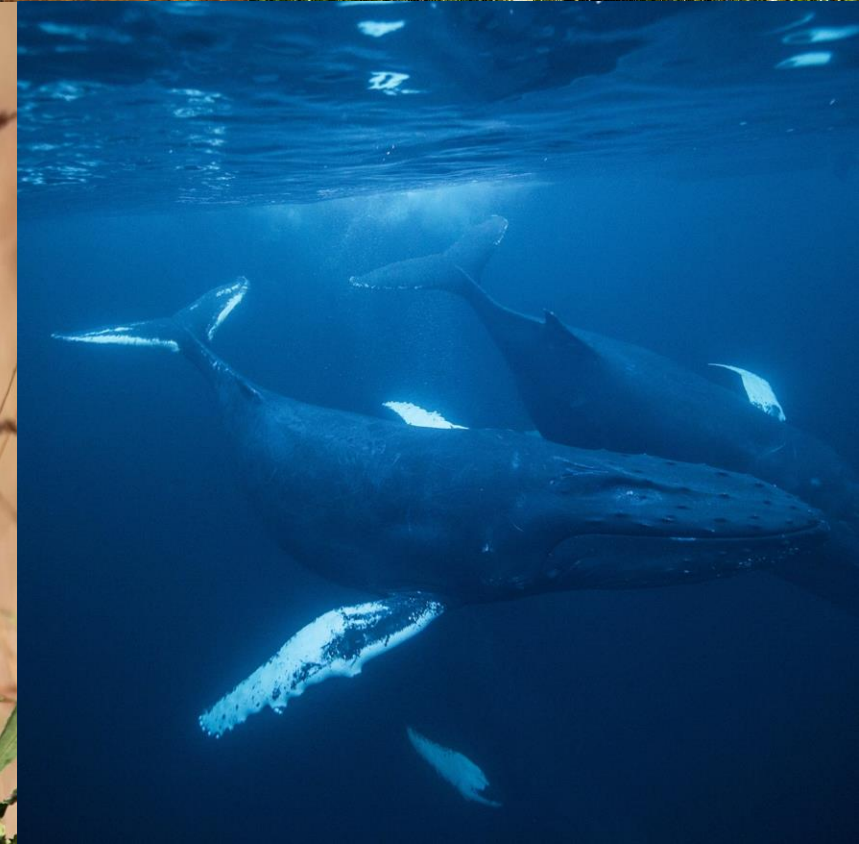




Table 2. Characteristics of the three search strategies and typical prey

<i>Behavioral patterns</i>	<i>Cruise strategy</i>	<i>Saltatory strategy</i>	<i>Ambush strategy</i>
Search			
When	while moving	while pausing	while pausing
Duration of pause	rarely pause	brief	long
Variation in duration of pause	rarely pause	related to prey characteristics and environmental conditions	related to prey characteristics and environmental conditions
Prey encounter			
When	while moving	while pausing	while pausing
Where in volume searched	outer boundary only	throughout	outer boundary only
Forward movement after pause			
Length of move	move continuously	related to size of prey	rarely move
Relationship to volume searched	none	length increases with size of area searched	uncertain
Relationship to prey density	length decreases with density of prey	none	none
Turning movement after pause			
Frequency	rare	after every pause	rare
When	while moving	only after pause	rarely turn
Relationship to volume searched	none	angle increases with size of area searched	none
Characteristics of prey			
Size relative to predator	large	small	large
Movement	variable	little	extensive
Major defense strategy	speed or hiding	hiding	large size

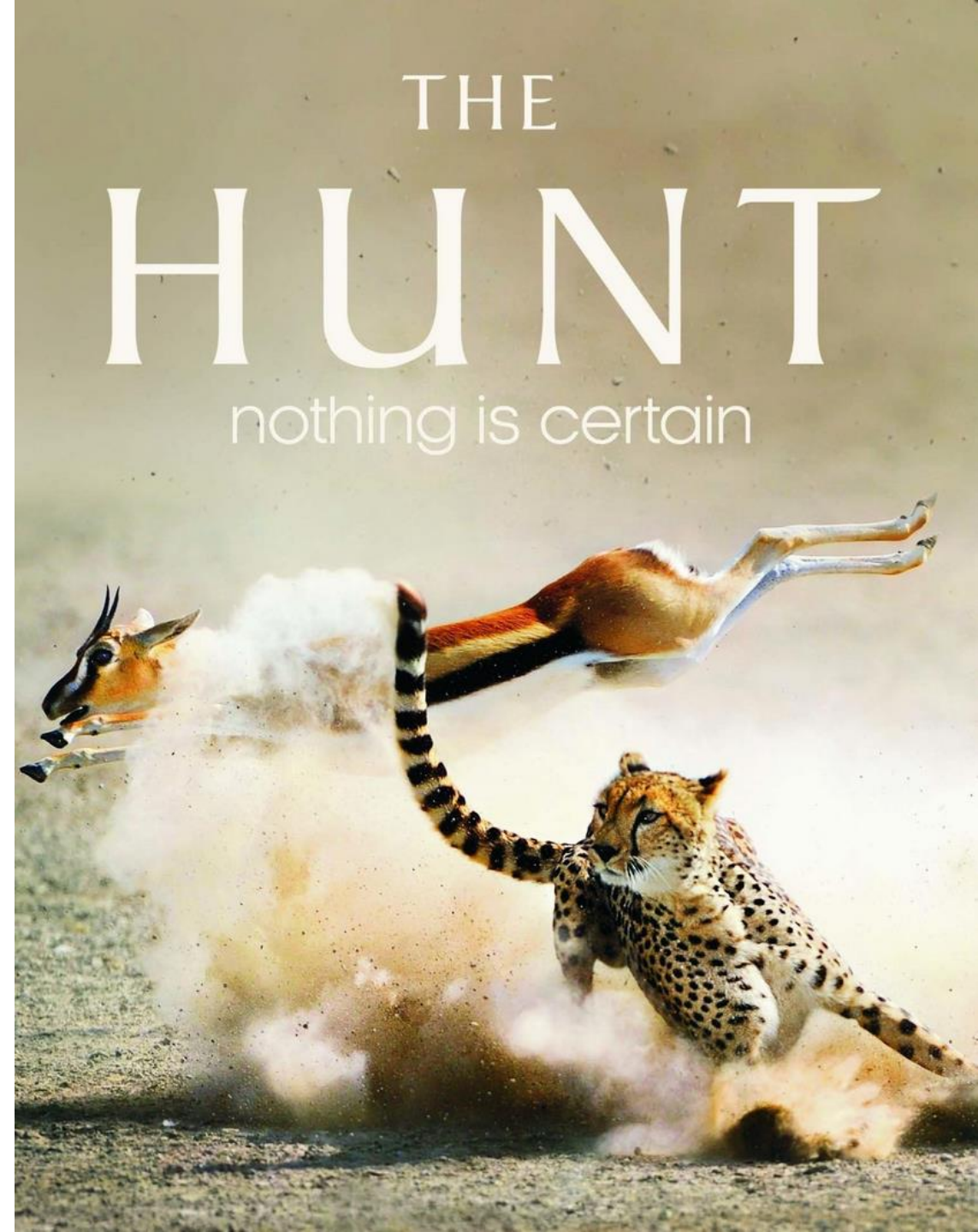
Handling

Typically what people think of when they hear foraging.

Food goes from being found to being consumed.

For predators, requires capturing and killing prey. For herbivores, may require overcoming defenses of plants.

Once obtained, prey needs to be consumed. May need to be broken down.



Alternate Foraging Strategies

Animals may adopt “alternate foraging strategies” or AFS.

Environmental change can spur adoption of AFS.

AFS can allow species to survive in rapidly changing environments.

Increasingly important in conservation studies.

Alternative foraging strategies enable a mountain ungulate to persist after migration loss

Alyson B. Courtemanch ✉, Matthew J. Kauffman, Steve Kilpatrick, Sarah R. Dewey

First published: 19 June 2017 | <https://doi.org/10.1002/ecs2.1855> | Citations: 20



Caching

- Not all food is immediately eaten.
- Some organisms cache– storing food for a later day.
- Return to cache and consume food after time has passed.
- Risks?
- Benefits?



Caching

- Food is perishable.
 - Cached food may go bad.
- Thieves.
- Organisms need to recall where they cached food.
 - Ever lost your keys?
- Increased upfront costs.
- Reduces future search time.
- Can account for seasonal fluctuations in resources.

Regular Article

Caching decisions by grey squirrels: a test of the handling time and perishability hypotheses

LEILA Z. HADJ-CHIKH ^{a f1}, MICHAEL A. STEELE ^{a f2}, PETER D. SMALLWOOD ^{b f3}



Group Foraging

If prey, increased vigilance against predators. If predator, can target larger/better defended prey.

Beneficial if you are a poor forager (old, young).

Can exploit ephemeral patches more easily.

Learning (such as in Aplin)

What are the costs and complications?



Group Foraging

- Communication (models assume species can truthfully communicate food resource information).
- Dominance (lower ranked individuals may receive less food)
- Increased visibility to potential rivals/predators.

Colin W. Clark, Marc Mangel. The evolutionary advantages of group foraging. *Theoretical Population Biology*, Volume 30, Issue 1, 1986, Pages 45-75.



Landscapes of Fear

Risk associated with habitat impacts foraging behavior.

Two requirements.

-- Variation in predation risk across habitat.

-- Ability for prey species to recognize and respond to this variation.

Originally proposed in the context of Yellowstone and wolf reintroduction.

Kohl, M.T., Stahler, D.R., Metz, M.C., Forester, J.D., Kauffman, M.J., Varley, N., White, P.J., Smith, D.W. and MacNulty, D.R. (2018), Diel predator activity drives a dynamic landscape of fear. *Ecol Monogr*, 88: 638-652. <https://doi.org/10.1002/ecm.1313>



Disturbance Foraging

Responder follow another organism (disturber) to obtain food.

Streamlines search period.

Certain bird species obtain >50% of food by following ant swarms.

Typically used by species that consume live prey (not usually conducted by herbivores).

Dominguez, J. S., Hauber, M. E., Tarwater, C. E., Williams, E., MacDonald, S., Strejc, B., & Pollock, H. S. (2023). Following the feeder: A global synthesis of disturbance-based foraging associations of birds. *Journal of Animal Ecology*, 92, 2263–2279. <https://doi.org/10.1111/1365-2656.14024>



Kleptoparasitism

Steal food from other species.

Can access food resources not always available to thief.

Can be time/energy intensive.

Similar to disturbance foraging, but explicitly antagonistic.





Kleptoparasitism

Kleptoparasitism may be opportunistic as opposed to specialized.

If specialized, requires a consistent host to steal from (meerkats and babblers for drongos, other seabirds for frigatebirds).

Kleptoparasitism can be boom-or-bust.

If hosts do badly, so will you.



Drongos

Mimics (as many as 43 unique calls).

Produce alarm calls to the benefit of other species (meerkats, squirrels, other birds).

Consistent kleptoparasites (at least 8 different drongo species regularly steal).

